Student Performance Analytics System

Project Documentation — Assessment Activity

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Course: Python Internship (EWB Courses) | **GitHub:** github.com/anshikapal2512-dora

1. Project Title

Project Name: Student Performance Analytics System | **Author:** Anshika Pal (anshikapal2512@gmail.com) | **Context:** EWB Courses Python Internship Submission

2. Objective

The project delivers an automated analytics application to track, evaluate, and extract meaningful academic insights from student records. It eliminates manual, error-prone calculation of totals, averages, percentages, and grades, while providing personalized student report cards and batch-level insights for educators.

3. Technologies Used

Technology / Tool	Role & Usage in Project
Python 3	Core language for application logic, calculations, and interactive loop.
Pandas	Dataframe manipulation, CSV reading, vectorized statistics (.mean, .idxmax, .sort_values).
NumPy	Numerical operations, vectorized mean evaluations, and statistical calculations.
OS Module	Dynamic relative path resolution ensuring portability across operating systems.
Git & GitHub	Version control, source code hosting, and project submission tracking.

4. Dataset Description

The dataset **students1.csv** contains 20 student entries indexed by Student_ID.

Column	Type	Description	Example
Student_ID	Integer	Unique student identifier (DataFrame index)	1
Name	String	Full name of the student	Aman
Department	String	Academic engineering department	Computer Engineering
English	Integer	Marks scored in English (out of 100)	78
Hindi	Integer	Marks scored in Hindi (out of 100)	67
Maths	Integer	Marks scored in Mathematics (out of 100)	56
Attendance	Float	Overall student attendance percentage	75.0

5. Implementation

1. **File Path Resolution:** Uses os.path.dirname(os.path.abspath(__file__)) to dynamically locate students1.csv, guaranteeing portability.
2. **Data Ingestion:** Loads the CSV using pd.read_csv(csv_path, index_col='Student_ID') and reports total enrolled records.
3. **Student Lookup & Evaluation:** Performs case-insensitive, whitespace-trimmed lookup. Calculates total marks, average, percentage, grade (A: ≥90, B: ≥75, C: ≥60, D: ≥35, F: <35), and pass/fail standing (≥35).
4. **Interactive Menu Loop:** Implements a continuous while True loop presenting 5 analytical choices, preventing accidental script exits.

6. Key Features

- **Case-Insensitive Search:** Handles name inputs in any casing with extra spaces stripped.
- **Automated Report Card Generation:** Computes totals, averages, grades, percentage, and pass/fail status.
- **Batch Statistical Analytics:** Instant identification of highest scorer, best subject, and class averages.

- **Persistent User Loop:** Users can execute multiple queries in one session before choosing to exit.
- **Portable Architecture:** Safe column access by name and portable relative path resolution.

7. Output / Execution Results

```
_____Student Performance Analytics System_____
Total number of students in the dataset: 20

Enter the name of the student to get the report: Aman
_____Student_Report_____
Student_ID: 1 | Name: Aman | Department: Computer Engineering
Marks -> English: 78 | Hindi: 67 | Maths: 56
Total: 201 | Average: 67.0 | Grade: C | Status: Pass | Percentage: 67.0% | Attendance: 75.0%

_____Questions/Answers Menu_____
[1] Highest Scorer: Kajal (Average: 87.00)
[2] Best Subject: Hindi (Average: 84.35)
[3] Passed Students: 20 out of 20 passed
[4] Overall Class Average: 79.70
```

8. Final Outcome

The project successfully achieved all specified goals. The Python CLI analytics application reliably reads and evaluates student records, generates complete individualized report cards, and provides instant batch-level metrics. The project is fully tested, documented, version-controlled with Git, and hosted on GitHub.

9. Challenges & Learning

- **Hardcoded Paths:** Windows absolute paths broke portability; solved via dynamic relative paths using os.path.
- **Case Sensitivity:** Searches for lowercase names previously failed; resolved with .str.strip().str.lower().
- **Repeated Computations:** Multiple mean calculations were optimized by caching values in local variables.
- **Single-Run Menu:** Replaced immediate script termination with an interactive while True loop.

10. GitHub Repository

Repository Link: https://github.com/anshikapal2512-dora/student-performance-analytics
Contains main.py, students1.csv, README.md, PROJECT_DOCUMENTATION.md, and Project_Documentation.pdf.